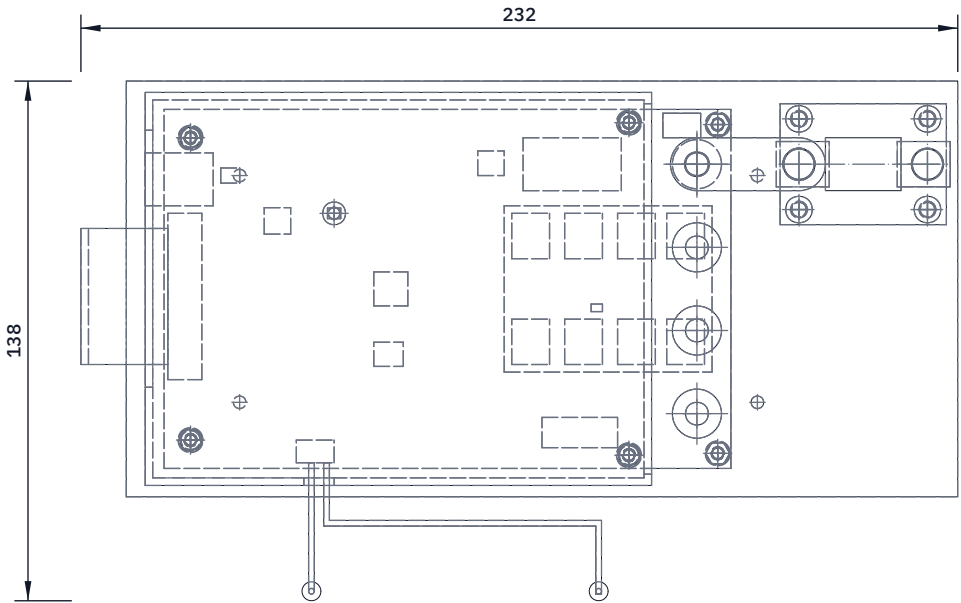


CONCEPT, NOT FOR FABRICATION

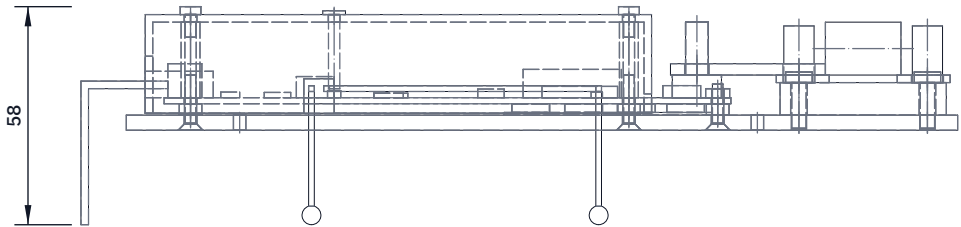
CellGuard · github.com/BoujeeEnjinia1701/cellguard

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA (CGD-CAL-001, CGD-DDR-001)	2026-09-25	AC
P2	Made buildable: fixings, fuse link, cover notches (CGD-DDR-003)	2026-09-30	AC
P3	Status light, light pipe, 1.5 mOhm switches (CGD-DEC-001)	2026-10-02	AC



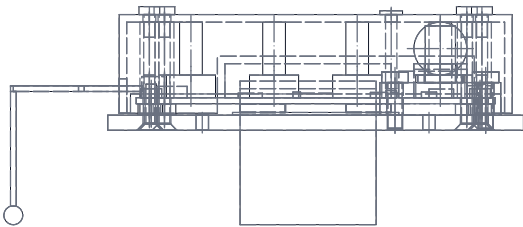
TOP VIEW

Scale 1:2; looking down (along -Z)



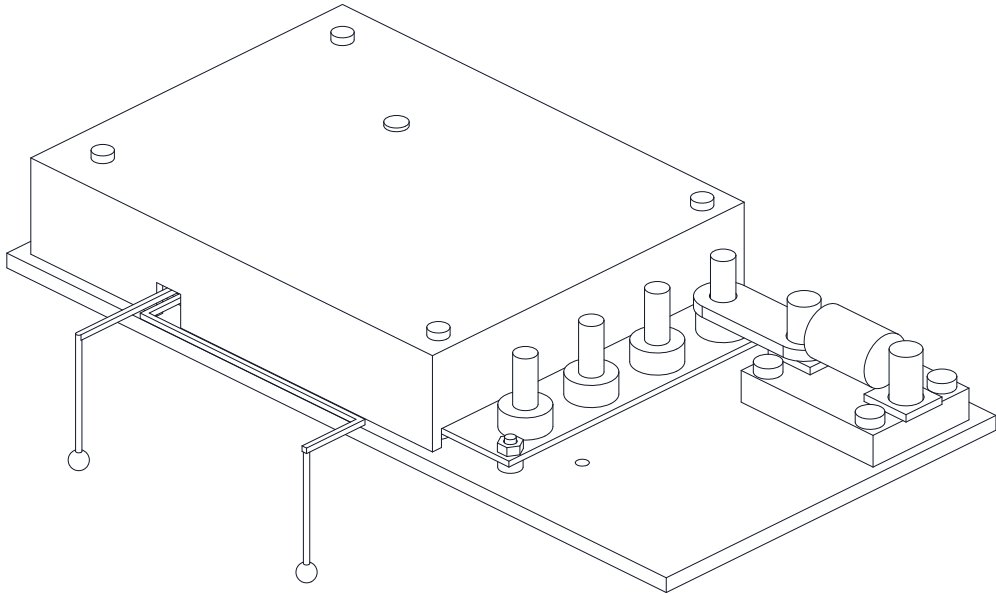
FRONT VIEW

Scale 1:2; looking at the front (along +Y)



RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS AND INTERFACES (MM)

Board envelope 220 x 110 x 33 (harness tail and probes extra)
Base plate 220 x 110 x 4, flat; board on 6 spacers 3 mm
PCB fixings: M3 countersunk from below; 4 pillars 20 mm carry the cover
PCB 150 x 95 x 1.6; top face at z 8.6
Power studs M6 at x 66: B-, P-, P+, B+ at y -33, -11, 11, 33
Pack fuse on plate in line with B+: 60 A, 80 V DC, 10 kA or more
Copper link 14 x 3 joins the B+ stud to the fuse holder
Plate mounting: 4 x M4 tapped, screws from below
Balance: 17-way connector at -X; CAN and UART 6-way at -X
MOSFETs 2 x 4, 1.5 mOhm class, under PCB on gap pad 1.0 to 0.7
Status light at x -30, y 20; 3 mm light pipe in the opaque cover
Secondary protector and SCP fuse in the B+ path (part 14)
CAN 2.0B 250 kbit/s, SwapCell v0.3 message set profile
40 A continuous, 80 A for 10 s; 4 to 16 LFP cells, under 60 V
Mass about 0.66 kg (CGD-CAL-001 v0.6)
PRELIMINARY, NOT FOR FABRICATION

CELLGUARD

github.com/BoujeeEnjinia1701/cellguard

General arrangement, BMS board

DRAWING NO. CGD-DWG-001			REV P3	SHEET 1 of 1	SIZE ANSI B
SCALE 1:2	UNITS mm	PROJECTION 	DATE 2026-10-02		DRAWN Amish Chadha
MATERIAL / NOTES Plate 6061 Al 4 mm; PCB FR-4 4-layer 2 oz; cover PC UL 94 V-0, opaque. See bom/bom.csv					
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